

# Comprehensive Characterization and Python-Based Data Analysis of Iron Oxide Nanoparticles

*A portfolio demonstration of a five-technique materials-characterization workflow built on synthetic datasets*

## DEMONSTRATION PROJECT — SYNTHETIC DATA

Every dataset analysed in this report was generated computationally by `scripts/generate_synthetic_datasets.py`. No material was synthesized, no instrument was operated, and no measurement was performed by the author or by anyone else in the production of this document. The purpose of the project is to demonstrate competence in processing, quantifying and critically interpreting materials-characterization data — not to report experimental findings. Nothing in this report may be cited as evidence about any real sample.

## Abstract

This report demonstrates a complete analytical workflow for the characterization of magnetite ( $\text{Fe}_3\text{O}_4$ ) nanoparticles across five techniques — X-ray diffraction, Fourier-transform infrared spectroscopy, UV-visible spectroscopy, scanning electron microscopy and thermogravimetric analysis. All five datasets are synthetic, generated from explicit physical models with realistic noise, and are labelled as such throughout. The workflow covers data loading, background and baseline correction, peak detection, profile fitting, quantitative extraction, statistical description and cross-technique reconciliation, implemented in Python with NumPy, pandas, SciPy and Matplotlib.

The synthetic XRD pattern was indexed to the cubic inverse-spinel structure, giving a refined lattice parameter of  $8.3962 \pm 0.0073 \text{ \AA}$  by Nelson-Riley extrapolation. Pseudo-Voigt profile fitting with instrumental-broadening correction returned a mean Scherrer crystallite size of  $11.1 \pm 0.5 \text{ nm}$ , while a Williamson-Hall analysis separating size from strain returned  $11.9 \pm 0.6 \text{ nm}$ . The FTIR dataset yielded 6 resolved bands, assigned to spinel Fe-O lattice modes and adsorbed water, with one band explicitly identified as an atmospheric artefact rather than a sample feature. Statistical analysis of 84 particle measurements gave a number-mean diameter of  $19.1 \text{ nm}$ . The TGA curve resolved three mass-loss events totalling 4.65 %.

Two results are reported as *negative* outcomes, and deliberately so. First, the SEM number-mean diameter exceeds the XRD coherent-domain size by a factor of 1.60; the report explains why this is expected rather than contradictory. Second, Tauc analysis of the UV-Vis dataset is shown to be indefensible for this material: four equally justifiable analysis choices yield apparent optical gaps spanning 0.64–3.00 eV, and no band-gap value is reported. The analysis pipeline is independently validated against the parameters used to generate the data; all 55 automated checks pass.

## 1. Introduction

Iron oxide nanoparticles, and magnetite in particular, are among the most widely studied nanomaterials in chemistry and materials science. Their combination of magnetic response, redox activity, low cost and biocompatibility places them in applications ranging from magnetic separation and catalysis to contrast agents and targeted drug delivery [11, 14]. In every one of those applications, performance depends on properties that cannot be read off a synthesis protocol: the crystalline phase actually formed, the size of the coherently diffracting domains, the size and shape of the physical particles, what is adsorbed on their surfaces, and how much of it there is.

No single technique answers those questions. X-ray diffraction reports on long-range order and says nothing about surface chemistry. Infrared spectroscopy reports on bonding and says nothing about particle size. Electron microscopy shows morphology but samples a vanishingly small fraction of the material. Thermal analysis quantifies volatiles but cannot identify them. Characterization is therefore an exercise in combining partial, differently-biased views of the same material — and in recognising when those views disagree for good reasons.

Magnetite is a particularly instructive case because it is genuinely difficult to characterise. Its lattice parameter (8.396 Å) differs from that of maghemite,  $\gamma\text{-Fe}_2\text{O}_3$  (8.346 Å), by only about 0.6 %, so the two phases produce diffraction patterns that are similar to within a few tenths of a degree in  $2\theta$  [10, 19]. Magnetite also oxidises to maghemite readily, so partially oxidised samples give intermediate lattice parameters. And because magnetite is a mixed-valence conductor rather than a wide-gap semiconductor [16], optical analyses routinely applied to semiconducting oxides are not transferable to it. Each of these difficulties is treated explicitly in this report.

## 2. Project Objective

This is a **portfolio demonstration**. Its objective is to show, on data of known provenance, the analytical capabilities that a client would be buying: the ability to take raw characterization output, process it defensibly, extract quantitative parameters with stated uncertainties, and interpret the result — including knowing when a number should not be reported at all.

The objective is explicitly *not* to present experimental findings. A demonstration built on synthetic data has one significant advantage over one built on real data: because the generative parameters are known, the analysis can be **validated**. Section 10 and the accompanying *scripts/validate.py* compare every extracted quantity against the value that was injected. A pipeline that cannot recover known inputs is not one whose output should be trusted on unknown inputs.

Concretely, the project demonstrates: construction of physically reasonable datasets from first-principles models; background and baseline estimation; peak detection and non-linear profile fitting; application of the Scherrer and Williamson–Hall relations with correct unit handling; defensible spectroscopic band assignment, including refusal to assign; descriptive and distributional statistics on a particle population; derivative thermogravimetry and boundary-independent step deconvolution; and reconciliation of size metrics that disagree.

## 3. Demonstration Dataset and Methodology

### 3.1 The hypothetical scenario

To keep the five datasets mutually consistent they were designed around a single hypothetical scenario. **This scenario was never carried out.** It exists only to constrain the data generation:

- **Material.**  $\text{Fe}_3\text{O}_4$  (magnetite), cubic inverse spinel, space group  $\text{Fd-}3\text{m}$  (No. 227).
- **Assumed route.** Surfactant-free aqueous co-precipitation of  $\text{Fe(II)/Fe(III)}$  chloride salts at a 1:2 molar ratio in excess  $\text{NH}_4\text{OH}$ , 80 °C, under  $\text{N}_2$ , followed by magnetic decantation, washing to neutral pH, and vacuum drying at 60 °C. This is the classic surfactant-free Massart-type co-precipitation [12].
- **No capping agent.** The scenario assumes an uncoated surface. This is a binding design constraint: it is why the synthetic TGA data contain no organic-decomposition step and the synthetic FTIR data contain no C–H or C=O bands. Had a capping agent been assumed, both datasets would have been built differently [13, 23].
- **Inert TGA atmosphere.** Nitrogen is assumed. In air, magnetite oxidises above roughly 200 °C with a theoretical mass *gain* of +3.45 %, which would partly mask the mass losses. The atmosphere is therefore

not a detail but a determinant of the curve's shape [26, 27].

### 3.2 How the datasets were generated

Each dataset was produced from an explicit forward model plus realistic noise, not by drawing curves. The XRD pattern is a sum of pseudo-Voigt reflections whose widths follow the combined Scherrer and microstrain relations, placed on a decaying background and sampled with Poisson counting statistics. The FTIR spectrum is a sum of Gaussian and Lorentzian absorbance bands on a curved baseline. The UV-Vis spectrum is the sum of a charge-transfer edge, an Urbach tail and a  $\lambda^{-n}$  turbidity term. The SEM table is a lognormal diameter population with weakly size-correlated shape descriptors. The TGA curve is a sum of logistic mass-loss steps with buoyancy drift and balance noise.

The generative parameters are recorded in *scripts/config.py* and are treated as ground truth for validation. A single random seed (20240517) makes every dataset reproducible.

| Technique | Synthetic acquisition conditions                                                           | Points      | Raw file                      |
|-----------|--------------------------------------------------------------------------------------------|-------------|-------------------------------|
| XRD       | Cu K $\alpha_1$ , 1.54060 Å, monochromated; 20–80° 2 $\theta$ ; 0.02° step; Bragg-Brentano | 3001        | xrd_data.csv                  |
| FTIR      | ATR (assumed); 4000–400 cm <sup>-1</sup> ; 1 cm <sup>-1</sup> step                         | 3601        | ftir_data.csv                 |
| UV-Vis    | Aqueous dispersion, 1 cm path (assumed); 200–900 nm; 1 nm step                             | 701         | uvvis_data.csv                |
| SEM       | Post-segmentation measurement table; no micrograph fabricated                              | 84 features | sem_particle_measurements.csv |
| TGA       | N <sub>2</sub> assumed; 10 °C min <sup>-1</sup> ; 25–800 °C; 0.5 °C step                   | 1551        | tga_data.csv                  |

**Table 1.** The five synthetic demonstration datasets. Acquisition conditions describe what the data were designed to imitate; they are simulation parameters, not instrument logs.

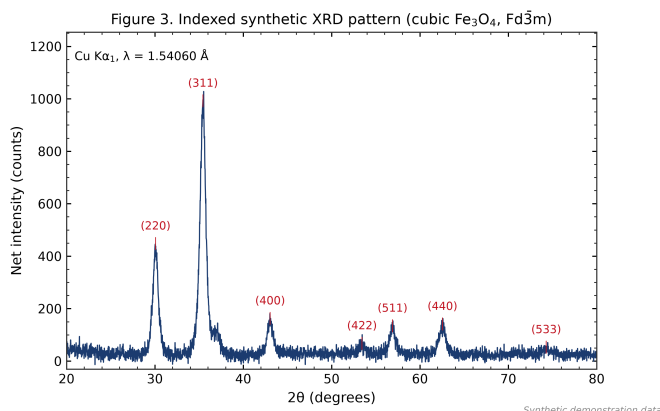
### 3.3 Software

The analysis uses Python 3.11 with NumPy [28], pandas [31], SciPy [29] and Matplotlib [30]. scikit-learn is installed but was deliberately not used: no step in this workflow is a machine-learning problem, and adding a model for appearance would have been dishonest about what the analysis does.

## 4. X-ray Diffraction Analysis

### 4.1 Processing and indexing

The raw synthetic pattern was background-corrected using an asymmetric least-squares (ALS) baseline [33, 34], which pulls a stiff baseline beneath the peaks rather than through them. Baseline stiffness matters more than it first appears: a baseline flexible enough to follow the Lorentzian peak wings artificially narrows the reflections and inflates the apparent crystallite size. Peak detection on the smoothed, background-subtracted trace located 11 candidate reflections, of which 7 were indexed against cubic Fe<sub>3</sub>O<sub>4</sub>.

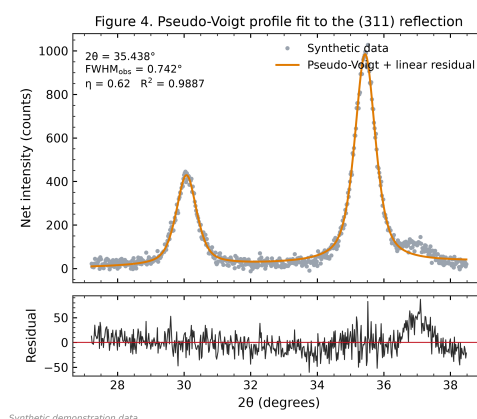


**Figure 3.** Background-subtracted pattern with Miller indices. The reflection sequence (220), (311), (400), (422), (511), (440) is the signature of the cubic inverse-spinel structure, space group Fd3m. The (311) reflection is the strongest, as expected.

### 4.2 Profile fitting

Each indexed reflection was fitted with a pseudo-Voigt profile — a linear combination of Gaussian and Lorentzian components sharing a common width [7] — plus a small linear residual background. Overlapping reflections were fitted simultaneously as a group. Two choices in this step have a direct effect on the extracted size, and both are worth stating:

- **Fit the background-subtracted pattern.** Fitting raw counts forces the local background parameters to trade off against the peak wings.
- **Scale the fitting window to each peak's own measured width.** A fixed window truncates the broad high-angle reflections more severely than the narrow low-angle ones. In an earlier iteration of this analysis a fixed  $\pm 1.1^\circ$  window produced fitted widths that were 3.9 % too large at  $30^\circ$  and 8.8 % too small at  $57^\circ$ , which tilted the Williamson–Hall slope far enough to yield a physically meaningless *negative* microstrain. Scaling the window to four times the measured half-width removed the artefact.



**Figure 4.** Pseudo-Voigt fit to the (311) reflection with residuals. Structureless residuals indicate that the profile model describes the peak shape adequately.

### 4.3 The Scherrer equation

Crystallite size was obtained from the Scherrer relation:

$$D = K \cdot \lambda / (\beta \cdot \cos \theta)$$

| Symbol    | Meaning                                                                            | Value used   | Unit handling                                                                                             |
|-----------|------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------|
| D         | Volume-weighted mean crystallite dimension perpendicular to the diffracting planes | computed     | returned in nm                                                                                            |
| K         | Shape factor; 0.9 is conventional for roughly spherical crystallites [2]           | 0.9          | dimensionless                                                                                             |
| $\lambda$ | X-ray wavelength, Cu $K\alpha_1$                                                   | 1.54060 Å    | converted to 0.154060 nm so that D comes out in nm                                                        |
| $\beta$   | Broadening of the reflection due to the sample alone                               | from the fit | <b>fitted FWHM in degrees must be converted to radians</b> ; omitting this inflates D by a factor of 57.3 |
| $\theta$  | Bragg angle                                                                        | from the fit | <b><math>\theta = 2\theta/2</math></b> ; the fitted position is $2\theta$ , the equation needs $\theta$   |

**Table 2.** Terms of the Scherrer equation and the two unit conversions that most commonly corrupt reported crystallite sizes. Both are verified explicitly by the validation suite.

Instrumental broadening was removed in quadrature,  $\beta_{\text{sample}} = \sqrt{(\beta_{\text{obs}}^2 - \beta_{\text{inst}}^2)}$ , with  $\beta_{\text{inst}} = 0.080^\circ$ . Averaged over the 5 reflections strong enough for their widths to be meaningful, this gives **D = 11.12 ± 0.46 nm**.

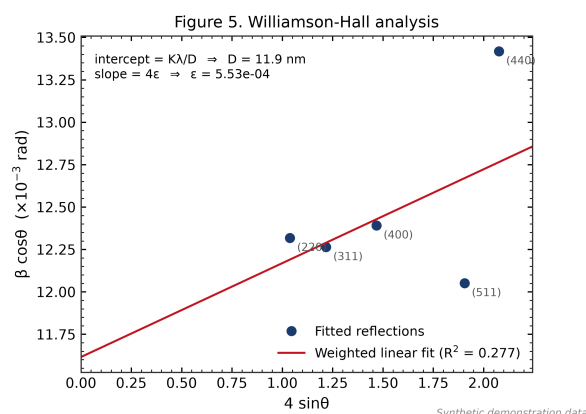
### 4.4 Separating size from strain

The Scherrer equation attributes *all* sample broadening to finite crystallite size. If the material also carries microstrain, that assumption biases the size low. The Williamson–Hall construction [4] separates the two contributions, since size broadening is constant in  $\beta \cdot \cos \theta$  while strain broadening grows as  $\sin \theta$ :

$$\beta \cdot \cos \theta = K \cdot \lambda / D + 4 \cdot \epsilon \cdot \sin \theta$$

Regressing  $\beta \cdot \cos \theta$  against  $4 \sin \theta$ , weighted by the uncertainty of each fitted width, gives an intercept corresponding to  $D = 11.94 \pm 0.59 \text{ nm}$  and a slope corresponding to  $\epsilon = 5.5\text{e-}04$ . As expected, the Williamson-Hall size exceeds the plain Scherrer size (11.12 nm), because the Scherrer treatment absorbs the strain contribution into the size term.

The fit quality,  $R^2 = 0.28$ , deserves comment rather than concealment. With only 5 usable reflections spanning a limited angular range, the *intercept* of a Williamson-Hall plot is far better constrained than the *slope*. The crystallite size is therefore reliable; the microstrain should be read as an order of magnitude ( $\sim 10^{-4}$ ), not as a precise value. Reporting  $\epsilon = 5.5\text{e-}04$  to two significant figures would overstate what these data support.



**Figure 5.** Williamson-Hall plot. The intercept yields the crystallite size and the slope the microstrain. Scatter about the line reflects the genuine difficulty of determining strain from a small number of broadened reflections.

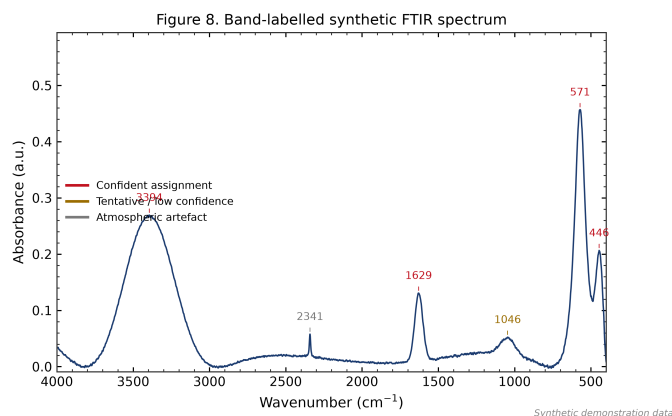
## 4.5 Phase identification and its limits

Nelson-Riley extrapolation [5, 9] of the peak-wise lattice parameters to  $\theta = 90^\circ$ , which cancels specimen-displacement and absorption errors, gives  $a = 8.3962 \pm 0.0073 \text{ Å}$ , using the 6 reflections whose positions are well determined. This sits within one standard error of the magnetite reference value (8.396 Å) and roughly seven standard errors from the maghemite value (8.346 Å), so for this dataset the assignment to  $\text{Fe}_3\text{O}_4$  is defensible.

That conclusion should not be over-generalised. The magnetite and maghemite lattice parameters differ by only 0.6 %, their reflections differ by 0.2–0.5° in  $2\theta$  across this range, and both are cubic spinels with nearly identical reflection sequences. On a real specimen — where partial surface oxidation, specimen displacement and zero-offset errors all shift peaks by comparable amounts — XRD peak positions alone are not sufficient to establish the phase [19]. A quantitative claim about magnetite content should be supported by Raman [21], Mössbauer or XPS evidence, or by Rietveld refinement of the full pattern [6] rather than single-peak fitting.

## 5. FTIR Analysis

The raw absorbance spectrum was corrected with a stiff ALS baseline to remove the smooth scattering offset typical of particulate sampling, then smoothed with an 11-point third-order Savitzky-Golay filter [32]. The smoothing width was chosen deliberately: the narrowest genuine feature in the dataset is about  $9 \text{ cm}^{-1}$  wide, and at  $1 \text{ cm}^{-1}$  sampling an 11-point filter has an effective width well below that, so band shapes and areas survive intact. Smoothing that is wide enough to look attractive is usually wide enough to distort the quantities being measured. Peak detection found 6 bands above  $3\sigma$  of the baseline noise ( $\sigma = 0.00634$  absorbance units).



**Figure 8.** Band-labelled synthetic FTIR spectrum. Colour encodes assignment confidence: confident, tentative, and atmospheric artefact. Treating all detected bands as equally interpretable is one of the commonest failures in applied vibrational spectroscopy.

| Band (cm <sup>-1</sup> ) | Assignment                                                            | Confidence | Basis and caveats                                                                                                                                                          |
|--------------------------|-----------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3394                     | O-H stretch: adsorbed molecular water + surface hydroxyls             | high       | Very broad -> extensive hydrogen bonding. Cannot distinguish H <sub>2</sub> O from structural OH on band position alone.                                                   |
| 2341                     | Atmospheric CO <sub>2</sub> asymmetric stretch -- INSTRUMENT ARTEFACT | high       | Not a sample feature. Arises from incomplete background compensation of ambient CO <sub>2</sub> .                                                                          |
| 1629                     | H-O-H bending of molecular water                                      | high       | Co-occurrence with the 3400 band supports molecular water rather than isolated OH groups.                                                                                  |
| 1046                     | Tentative: Fe-OH deformation / surface hydroxyl                       | low        | Weak and broad. Overlaps the region where residual sulfate or silicate would also absorb. NOT a confident assignment.                                                      |
| 571                      | Fe-O stretch, tetrahedrally coordinated Fe                            | high       | Principal spinel lattice mode. Position is phase-sensitive but NOT sufficient alone to separate Fe <sub>3</sub> O <sub>4</sub> from gamma-Fe <sub>2</sub> O <sub>3</sub> . |
| 446                      | Fe-O stretch, octahedrally coordinated Fe                             | medium     | Close to the low-wavenumber cut-off; intensity and position are less reliable here.                                                                                        |

**Table 3.** Detected bands with assignments, graded by confidence. Band positions follow Cornell & Schwertmann [10] and Gotić & Musić [22]; general assignment practice follows Stuart [20].

## 5.1 What can and cannot be concluded

The two strong low-wavenumber bands are the diagnostic spinel Fe-O lattice modes, conventionally associated with tetrahedrally and octahedrally coordinated iron [10, 22]. Their presence is consistent with a spinel iron oxide. It does *not* discriminate magnetite from maghemite: both are spinels, both absorb in this region, and reported band positions for the two overlap once particle size and sampling geometry are taken into account.

The broad feature near 3400 cm<sup>-1</sup> and the band near 1630 cm<sup>-1</sup> together indicate molecular water rather than isolated hydroxyl groups — the bending mode requires an H-O-H unit, so its co-occurrence with the stretch is what makes the assignment safe. A stretching band alone would not have justified it.

Two further points concern restraint. The weak feature near 1045 cm<sup>-1</sup> is reported as *tentative*: it lies in a region where surface Fe-OH deformation, residual sulfate and silicate contamination all absorb, and the data do not distinguish between them. Assigning it confidently would not be defensible. Separately, the sharp band near 2340 cm<sup>-1</sup> is identified as atmospheric CO<sub>2</sub> — an instrumental artefact from imperfect background compensation, not a property of the sample. It was placed in the synthetic data on purpose, to demonstrate that recognising artefacts is part of the analysis.

## 5.2 Evidence from absence

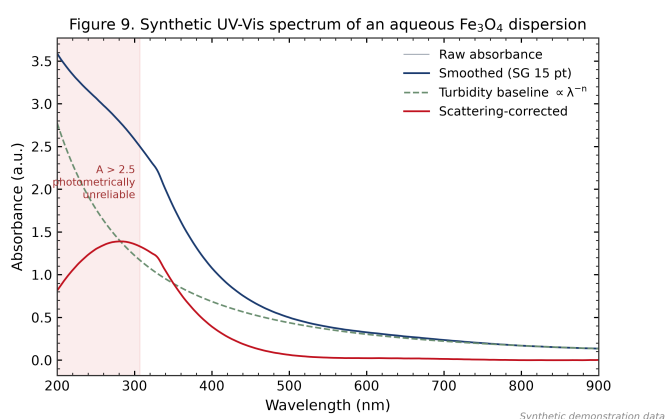
Bands that are *not* present carry information too, provided absence is tested properly. Asking merely whether absorbance is low in some window is unreliable, because the tail of a strong neighbouring band can fail that test — the 1630 cm<sup>-1</sup> water bend bleeds into the region where a carbonyl would appear. The defensible test is whether a *resolved local maximum* exists. On that basis:

- No C-H stretching bands ( $2840\text{--}2970\text{ cm}^{-1}$ ) and no carbonyl band ( $\sim 1700\text{ cm}^{-1}$ ) are resolved, consistent with the design assumption of an uncoated surface and with the absence of an organic step in the TGA data.
- No bands at  $\sim 890$  and  $\sim 795\text{ cm}^{-1}$ , where goethite ( $\alpha\text{-FeOOH}$ ) absorbs, arguing against that common co-precipitation impurity [10].
- No nitrate ( $\sim 1384\text{ cm}^{-1}$ ) or sulfate ( $\sim 1100\text{ cm}^{-1}$ ) bands, consistent with the assumed chloride precursors and adequate washing.

These are consistency checks, not proof: FTIR sensitivity to a trace phase is limited, and a few percent of goethite could be present without producing a resolved band.

## 6. UV-Visible Spectroscopy

The synthetic spectrum rises monotonically from the near-infrared into the ultraviolet with no discrete excitonic maximum — absorbance 0.33 at 600 nm rising to 1.08 at 400 nm. This broad, featureless profile is the optical signature of  $\text{O}(2p) \rightarrow \text{Fe}(3d)$  charge transfer together with a strong sub-edge tail, and it is why magnetite appears black rather than coloured [20 nm-class dispersions: 15].



**Figure 9.** Synthetic UV-Vis spectrum with the fitted turbidity baseline and the scattering-reduced curve. The shaded region marks absorbances above 2.5, where stray light makes a conventional spectrophotometer unreliable; those points are excluded from all subsequent analysis.

### 6.1 Two problems before any analysis begins

**Photometric range.** 107 of the 701 points exceed absorbance 2.5, meaning the data below roughly 307 nm are compromised by stray light. They are flagged and excluded rather than fitted. Extrapolating a Tauc line through saturated data is a common and undetectable error.

**Scattering.** For a dispersion, what a spectrophotometer records is not absorption but attenuation — absorption plus turbidity. Fitting an empirical power law over 780–900 nm gives an effective exponent of  $n = 2.02$ , and the scattering term accounts for roughly 93 % of the measured absorbance at 600 nm. Note that this exponent is *effective*, not a clean Rayleigh or Mie value: genuine absorption persists across the fitting window, so the power law unavoidably absorbs some of it. The corrected spectrum is best described as turbidity-reduced, not scattering-free.

### 6.2 Tauc analysis, and why no band gap is reported

The Tauc relation [17] expresses the absorption edge of a semiconductor as  $(\alpha \cdot h\nu)^{1/r} = B(h\nu - E_g)$ , with  $r = \frac{1}{2}$  for a direct allowed transition and  $r = 2$  for an indirect allowed one. Extrapolating the linear portion to the energy axis gives the apparent gap. The method is demonstrated here in full — and then **not used to report a result**. Three independent reasons:

- **The material is wrong for the method.** Magnetite is a mixed-valence, near-metallic conductor above the Verwey transition [16]; it does not possess the well-defined optical gap the Tauc model presumes. The relation was derived for amorphous semiconductors [17].
- **The quantity is wrong.** The Tauc relation requires the absorption coefficient  $\alpha$ . What is available is the absorbance of a dispersion, which conflates absorption with scattering and depends on an unknown effective path length and concentration.

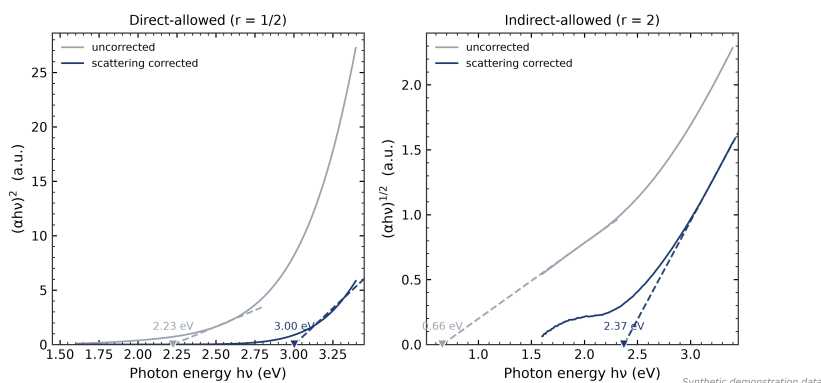


- **The answer is not stable.** This is the decisive point, and it is quantified below.

| Baseline treatment   | Assumed transition | Apparent $E_g$ (eV) | Fit $R^2$ |
|----------------------|--------------------|---------------------|-----------|
| uncorrected          | direct allowed     | 2.23                | 0.9929    |
| uncorrected          | indirect allowed   | 0.66                | 0.9999    |
| scattering corrected | direct allowed     | 3.00                | 0.9758    |
| scattering corrected | indirect allowed   | 2.37                | 0.9998    |

**Table 4.** Apparent optical gaps from four equally defensible analysis choices applied to the same dataset. Every fit has  $R^2 \geq 0.97$  — the fits all look excellent — yet the values disagree by 2.34 eV. Allowing additionally for the analyst's choice of linear region, the full defensible range is 0.64–3.00 eV.

Figure 10. Tauc analysis shown as a METHOD DEMONSTRATION -- the extracted values are not a band gap of  $\text{Fe}_3\text{O}_4$



**Figure 10.** Tauc plots for both transition assumptions, with and without scattering correction. The four extrapolations are individually convincing and mutually irreconcilable. This figure is presented as a cautionary method demonstration.

A spread of 2.36 eV is larger than any difference such an analysis would typically be trying to resolve. Reporting a single value from this dataset — say '2.37 eV', quoting its excellent  $R^2$  — would be indefensible, yet it is exactly what routinely appears in the literature. Makuła and co-workers [18] document this failure mode in detail. The correct output of this section is a refusal to report a number, together with the evidence for that refusal.

More generally, optical properties measured on a dispersion should not be treated as intrinsic electronic properties. They are convolved with particle size and aggregation state, which control scattering; with defect and oxidation state, which alter sub-edge absorption; with the solvent and surface chemistry; and with the concentration and path length of the measurement. Separating an intrinsic property from that convolution requires diffuse-reflectance measurement on a solid with Kubelka-Munk treatment, not a cuvette spectrum.

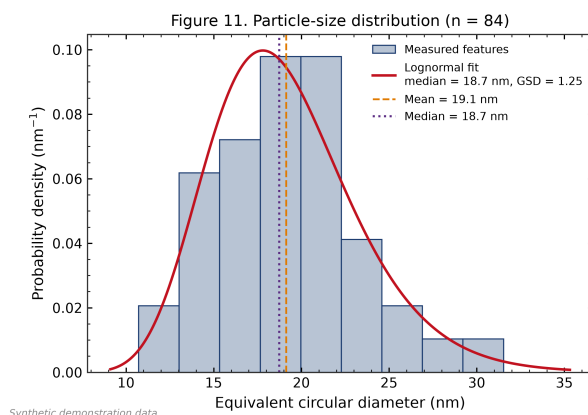
## 7. SEM-Based Particle-Size Analysis

**No micrograph is presented in this report.** Fabricating an image and describing it as an electron micrograph would be misrepresentation, regardless of any disclaimer. What is analysed instead is a table of 84 particle measurements of the kind produced *after* segmenting a real micrograph in software such as ImageJ/Fiji — the numeric input to the statistical workflow, which is the part being demonstrated.

| Statistic                       | Value            | Statistic               | Value    |
|---------------------------------|------------------|-------------------------|----------|
| n                               | 84               | Minimum                 | 10.72 nm |
| Mean                            | 19.13 nm         | Q1                      | 16.24 nm |
| Median                          | 18.74 nm         | Q3                      | 21.48 nm |
| Standard deviation              | 4.19 nm          | Maximum                 | 31.53 nm |
| Standard error                  | 0.457 nm         | IQR                     | 5.25 nm  |
| Coefficient of variation        | 21.9 %           | Skewness                | 0.478    |
| 95 % CI on the mean (bootstrap) | 18.24 – 20.05 nm | Excess kurtosis         | 0.377    |
| Volume-weighted D[4,3]          | 21.87 nm         | Surface-weighted D[3,2] | 20.95 nm |
| Mean aspect ratio               | 1.091            | Mean circularity        | 0.936    |



**Table 5.** Descriptive statistics of the synthetic particle population. The confidence interval is a 10 000-resample bootstrap, which avoids assuming normality — appropriate for a right-skewed distribution.



**Figure 11.** Size distribution with a fitted lognormal density. Mean and median are marked separately; their separation is the signature of right skew.

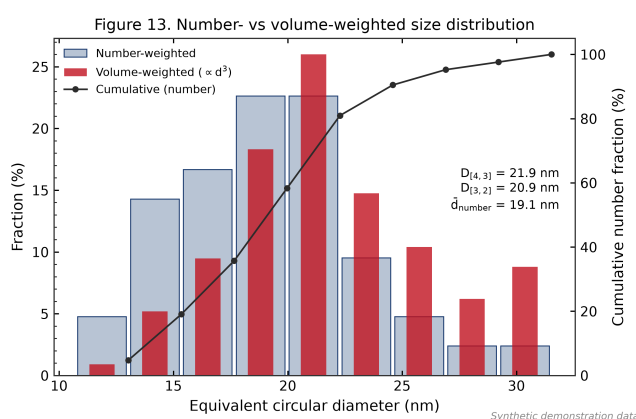
## 7.1 Distribution shape — and the limits of $n = 84$

The distribution is right-skewed (skewness 0.48), which is characteristic of populations formed by nucleation and growth and is conventionally modelled as lognormal. A maximum-likelihood lognormal fit gives a median of 18.68 nm and a geometric standard deviation of 1.245.

Honesty requires a caveat that is usually omitted. A Kolmogorov-Smirnov test does not reject the lognormal model ( $p = 0.755$ ) — but neither does it reject a *normal* model ( $p = 0.842$ ). With 84 measurements these data simply lack the power to discriminate between the two. The lognormal fit is therefore reported as a reasonable description justified by the physics of particle formation, not as a distribution demonstrated by these data. Claiming the latter would be a misuse of a non-significant test result.

## 7.2 Number weighting versus volume weighting

A raw arithmetic mean of diameters is *number-weighted*: every particle counts once. Many physical properties are not. Scattering, sedimentation and X-ray diffraction all weight toward larger particles, which is why the volume-weighted De Brouckere mean  $D[4,3] = \Sigma d^4 / \Sigma d^3 = 21.87$  nm exceeds the number mean of 19.13 nm, and the surface-weighted Sauter mean  $D[3,2] = 20.95$  nm falls between them. Comparing a number-weighted SEM mean directly against a volume-weighted XRD size — as is done constantly — compares two different statistics of two different quantities [24].



**Figure 13.** Number- and volume-weighted distributions of the same population, with the cumulative number fraction. The volume weighting visibly shifts the distribution toward larger sizes.

## 7.3 What image-derived sizes cannot tell you

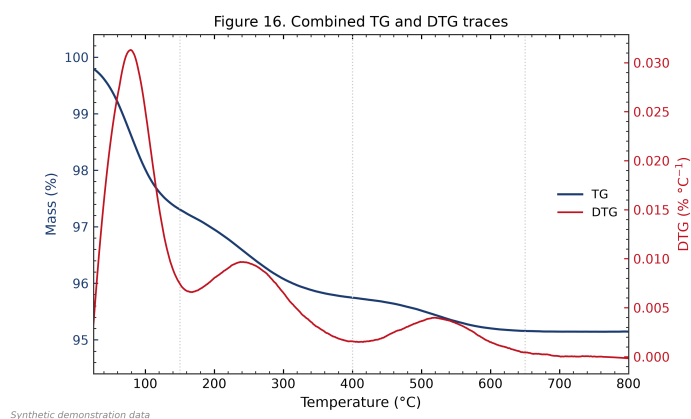
- **Agglomeration.** Automated segmentation frequently merges touching particles into one feature. The measured population is therefore of *features*, not necessarily of primary particles [25].
- **Thresholding.** The measured diameter depends on where the particle edge is judged to lie. Reasonable threshold choices routinely shift a mean diameter by several percent — usually more than the quoted

standard error.

- **Sampling bias.** A micrograph covers a minute fraction of a specimen, and fields of view are chosen by an operator who tends to select clear, well-dispersed, representative-looking regions.
- **Field-of-view truncation.** Particles clipped by the frame edge are typically discarded, which preferentially removes large ones.
- **Coating and edge effects.** A conductive coating adds real thickness, and edge-brightness effects broaden apparent outlines. Both bias diameters upward [25].
- **Sample size.** With  $n = 84$ , the 95 % confidence interval on the mean is 18.2-20.1 nm. Distribution parameters are far less certain than that, and tail behaviour is essentially unconstrained.

## 8. Thermogravimetric Analysis

The synthetic TGA curve was smoothed with a Savitzky-Golay filter and differentiated analytically from the same local polynomial to give the DTG trace,  $-dm/dT$ . Computing a derivative by finite differences on noisy data amplifies noise and generates spurious maxima; a wider window is used for the derivative than for the mass curve itself for exactly that reason. Total mass loss over 25–800 °C is **4.65 %**, leaving a residue of 95.14 %.



**Figure 16.** Combined TG and DTG traces. DTG maxima locate the temperature of fastest mass loss and resolve events that appear as a single gradual slope on the TG curve alone.

| Region     | Range (°C) | Mass loss (%) | % of total | DTG max (°C) | Assigned process                                                                                                               | Confidence |
|------------|------------|---------------|------------|--------------|--------------------------------------------------------------------------------------------------------------------------------|------------|
| Region I   | 25–150     | 2.49          | 53.5       | 79           | Loss of physisorbed and interparticle water                                                                                    | high       |
| Region II  | 150–400    | 1.56          | 33.6       | 238          | Loss of strongly bound water and surface dehydroxylation ( $2 \text{ Fe-OH} \rightarrow \text{Fe-O-Fe} + \text{H}_2\text{O}$ ) | medium     |
| Region III | 400–650    | 0.59          | 12.7       | 519          | Residual dehydroxylation / loss of lattice-associated OH                                                                       | low        |
| Region IV  | 650–800    | 0.01          | 0.3        | 651          | Comparatively stable; no further resolvable mass loss                                                                          | high       |

**Table 6.** Region-resolved mass losses. Confidence grades reflect that TGA measures mass change only: the process assignments are inferences from temperature range and from the FTIR evidence for adsorbed water, not direct observations.

### 8.1 Region integration versus step deconvolution

Region-based integration has an unavoidable weakness: the answer depends on where the analyst places the boundaries, and adjacent events overlap across them. Fitting the entire curve as a sum of logistic steps plus a linear drift term removes that arbitrariness and returns the amplitude of each *event* rather than of each temperature interval. The fit ( $R^2 = 1.00000$ ) resolves 3 events:

| Event  | Midpoint (°C) | Width (°C) | Amplitude (%)   |
|--------|---------------|------------|-----------------|
| Step 1 | 78.0          | 21.1       | $2.60 \pm 0.00$ |
| Step 2 | 243.1         | 43.9       | $1.70 \pm 0.00$ |

| Event  | Midpoint (°C) | Width (°C) | Amplitude (%) |
|--------|---------------|------------|---------------|
| Step 3 | 521.8         | 38.1       | 0.60 ± 0.00   |

**Table 7.** Boundary-independent step deconvolution. Summed amplitude 4.90 % slightly exceeds the measured total loss of 4.65 %, because part of the lowest-temperature event has already occurred before the run reaches its 25 °C starting point — a real effect that causes TGA to under-report total volatile content.

## 8.2 Interpretation

The assignment of these events rests on the combination of temperature range and independent evidence. Loss below roughly 150 °C is conventionally physisorbed and interparticle water; loss between about 150 and 400 °C is attributed to more strongly bound water and to surface dehydroxylation,  $2 \text{ Fe-OH} \rightarrow \text{Fe-O-Fe} + \text{H}_2\text{O}$ ; the small high-temperature event is consistent with residual dehydroxylation [10, 23]. Crucially, the FTIR spectrum independently shows water and hydroxyl species and shows no organic bands, which is what licenses a water-based interpretation rather than an organic one. TGA alone could not distinguish them.

Combining techniques makes the interpretation quantitative. Treating the crystallites as spheres of the Williamson–Hall diameter (11.9 nm) gives a specific surface area of about  $97 \text{ m}^2 \text{ g}^{-1}$ . Converting the Region II mass loss into molecules over that area gives roughly **5.4 H<sub>2</sub>O molecules nm<sup>-2</sup>**. That falls within the range generally quoted for hydroxylated iron-oxide surfaces (about 2–12 nm<sup>-2</sup>) [10, 23], which is a genuine internal consistency check: the diffraction, thermal and spectroscopic datasets tell a compatible story. The figure is an order-of-magnitude estimate — it inherits the sphericity assumption, ignores porosity and agglomeration, and would change if the true surface area differed.

## 8.3 Limitations of thermal interpretation

- **TGA measures mass, not identity.** Every process assignment above is an inference. Only a coupled technique — TGA-MS or TGA-FTIR of the evolved gas — identifies what is actually leaving.
- **Atmosphere is decisive.** The inert atmosphere here is an *assumption*. In air, magnetite oxidises to maghemite and then hematite above roughly 200 °C, a mass *gain* of up to +3.45 % that would partly cancel the losses and could make a genuinely hydrated sample appear nearly anhydrous. A TGA result reported without its atmosphere is uninterpretable [27].
- **Heating rate shifts everything.** Faster heating displaces DTG maxima to higher temperatures and merges adjacent events. The  $10 \text{ °C min}^{-1}$  rate here is assumed, and quoted transition temperatures are meaningful only alongside it [26].
- **Sample mass, packing and crucible matter.** Deep beds impede gas escape and shift events upward; crucible material can catalyse decomposition.
- **Buoyancy is an artefact, not a signal.** Gas density falls as the furnace heats, producing an apparent mass gain of order 0.05 %. It is modelled explicitly here; in practice it requires a blank subtraction.

## 9. Cross-Technique Discussion

| Technique     | What it probes                                                                      | Quantitative output                                                                              | Principal limitation                                                                                                                                                                                                                             |
|---------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>XRD</b>    | Crystal structure, phase identity, lattice parameter, crystallite size, microstrain | d-spacings; $a$ = lattice parameter (Å); $D_{\text{Scherrer}}$ (nm); microstrain $\epsilon$ s    | Volume-weighted and blind to amorphous or poorly-ordered material; cannot separate Fe <sub>3</sub> O <sub>4</sub> from gamma-Fe <sub>2</sub> O <sub>3</sub> on peak position alone; Scherrer size is a coherent-domain size, NOT a particle size |
| <b>FTIR</b>   | Fe-O lattice modes; adsorbed water; surface hydroxyls; presence/absence of organics | Band positions (cm <sup>-1</sup> ); relative intensities; band areas (semi-quantitative at best) | Heavily overlapped bands in the Fe-O region; absorptivities are unknown so intensities are not concentrations; sampling geometry (ATR vs KBr) shifts band positions and intensities                                                              |
| <b>UV-Vis</b> | Charge-transfer absorption; qualitative optical behaviour; dispersion stability     | Absorbance vs wavelength; apparent Tauc intercepts (unreliable for this material)                | For a dispersion, measured absorbance mixes true absorption with Mie/Rayleigh turbidity; Fe <sub>3</sub> O <sub>4</sub> is not a wide-gap semiconductor, so a Tauc 'band gap' is not a meaningful intrinsic property                             |

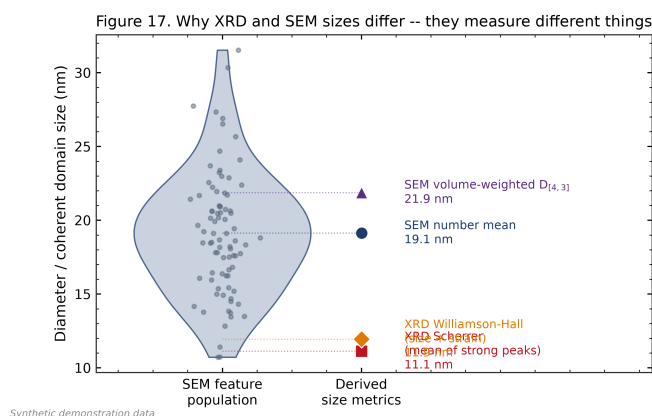
| Technique | What it probes                                                                                | Quantitative output                                                             | Principal limitation                                                                                                                                                                                                    |
|-----------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEM       | Particle/aggregate size, shape, degree of agglomeration, surface texture                      | Size distribution statistics: mean, median, SD, quartiles, D[4,3], D[3,2]       | Measures whole features, so aggregates are counted as particles; thresholding choices shift the mean by several percent; limited field of view creates sampling bias; conductive coating inflates diameters             |
| TGA       | Volatile content: adsorbed water, surface hydroxyls, organics (if present); thermal stability | Mass loss per region (%); DTG peak temperatures (C); total volatile content (%) | Gives mass change only, never chemical identity; results depend strongly on atmosphere, heating rate, sample mass and packing; in air, Fe <sub>3</sub> O <sub>4</sub> oxidation causes a mass GAIN that can mask losses |

**Table 8.** What each technique contributes and where it fails. The limitation column is the operative one: it is what determines which techniques must be combined.

## 9.1 Reconciling the XRD and SEM size metrics

The two size measurements disagree, and the disagreement is the most informative single result in this project. The SEM number-mean diameter (19.1 nm) exceeds the XRD coherent-domain size (11.9 nm) by a factor of 1.60. Even comparing like with like, on a volume-weighted basis, the factor is still 1.83. This is expected, not contradictory, and treating it as an error would be the mistake:

- **Different measurands: XRD reports the size of coherently diffracting domains; SEM reports the outline of a visible feature.** A single SEM feature may contain several domains.
- **Different weightings: Scherrer sizes are volume-weighted while a raw SEM mean is number-weighted.** Volume weighting emphasises the largest particles, so the two statistics are not comparable until one is converted (here D[4,3]).
- **Surface disorder: a structurally disordered or hydrated surface shell contributes to the SEM outline but diffracts incoherently, so XRD does not count it.**
- **Agglomeration: touching particles are frequently segmented as one feature, inflating the SEM size.**
- **Microstrain: if strain broadening is attributed entirely to size, the Scherrer size is systematically UNDERESTIMATED.** Here Scherrer alone gives 11.1 nm while the Williamson-Hall separation gives 11.9 nm.
- **Imaging artefacts: conductive coating thickness and edge detection thresholds both bias SEM diameters upward.**

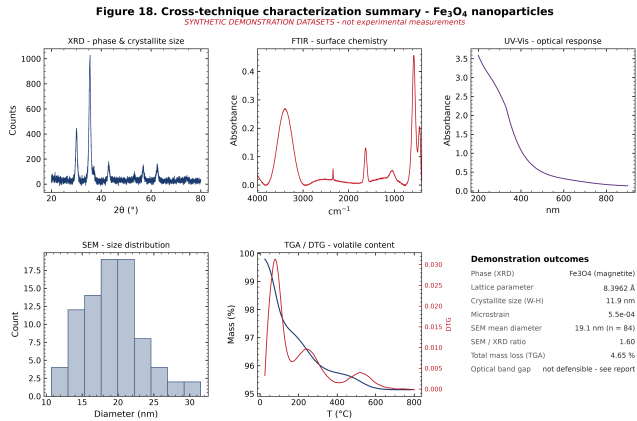


**Figure 17.** The SEM feature population against the derived size metrics. The techniques are not in conflict; they measure different physical quantities with different statistical weightings.

## 9.2 Complementarity

None of these techniques is a substitute for another. XRD establishes what the crystalline phase is; SEM establishes how the material is physically organised; FTIR reports what sits on the surface; TGA quantifies how much of it there is; UV-Vis reports the optical consequence. A defensible characterisation requires the set, and phase assignment in particular should be confirmed by a magnetic or spectroscopic method (Mossbauer, Raman, XPS) rather than by XRD peak positions alone.

The five techniques answer five different questions, and none substitutes for another. Crystallinity (XRD) tells you what phase formed but is blind to amorphous material and to everything on the surface. Morphology (SEM) shows how the material is physically organised but samples almost none of it. Surface chemistry (FTIR) identifies what is adsorbed but cannot say how much. Thermal behaviour (TGA) quantifies how much but cannot say what. Optical response (UV-Vis) reports a property that is itself a convolution of size, aggregation and electronic structure. The interpretation of the TGA water content in Section 8.2 required the XRD crystallite size and the FTIR band assignments to be meaningful at all — which is the general pattern, not an exception.



**Figure 18.** Single-page summary of all five techniques and the headline demonstration outcomes.

## 10. Scientific Limitations

### 10.1 The overriding limitation

**These data are synthetic.** They were generated from models the author wrote, and they therefore contain exactly the physics those models encode and no more. Real specimens produce complications this project cannot reproduce: preferred orientation, anisotropic crystallite shapes, stacking faults, amorphous fractions invisible to diffraction, inhomogeneity between aliquots, instrument drift, and sample-preparation artefacts. Nothing here constitutes evidence about any real material. The project demonstrates analytical method, not experimental findings.

### 10.2 Validation of the pipeline

Because the generative parameters are known, the analysis can be checked against them — something impossible with real data. *scripts/validate.py* runs 55 automated checks in three classes: recovery of injected parameters, internal consistency of units and arithmetic, and synthetic-data integrity. All 55 pass. The recovery results are the substantive ones:

| Quantity                           | Injected | Recovered         | Deviation |
|------------------------------------|----------|-------------------|-----------|
| Crystallite size (Williamson–Hall) | 12.00 nm | 11.94 nm          | -0.5 %    |
| Lattice parameter                  | 8.3960 Å | 8.3962 ± 0.0073 Å | +0.2 mÅ   |
| Microstrain                        | 8.0e-04  | 5.5e-04           | -31 %     |
| Lognormal median (SEM)             | 18.50 nm | 18.68 nm          | +1.0 %    |
| Geometric SD (SEM)                 | 1.280    | 1.245             | -2.7 %    |
| TGA step 1 amplitude               | 2.60 %   | 2.60 %            | +0.00 pp  |
| TGA step 2 amplitude               | 1.70 %   | 1.70 %            | -0.00 pp  |
| TGA step 3 amplitude               | 0.60 %   | 0.60 %            | +0.00 pp  |

**Table 9.** Recovery of injected parameters. The microstrain is the weakest recovery, for the reason given in Section 4.4: the Williamson–Hall slope is poorly constrained by a handful of reflections. It is reported as an order of magnitude rather than a value.

### 10.3 Method-specific limitations

- **Scherrer analysis.** Returns a volume-weighted mean of coherent domain size, not a particle size. It assumes strain-free crystallites unless corrected, a shape factor that is genuinely uncertain at the  $\pm 10\%$  level, and correct removal of instrumental broadening. It becomes unreliable above roughly 100 nm,

where size broadening falls below instrumental resolution [3].

- **Single-peak fitting.** Rietveld refinement of the whole pattern [6] uses all reflections simultaneously and would give better-determined parameters; it was not used here because the point was to demonstrate the underlying single-peak methods explicitly.
- **$K\alpha_2$ .** The synthetic pattern was generated for a single wavelength, as if monochromated or  $K\alpha_2$ -stripped. Raw laboratory data contain the doublet, which broadens and skews reflections at high angle and must be removed before width analysis.
- **FTIR quantification.** Band intensities are not concentrations without known absorptivities. ATR and KBr sampling give different relative intensities and slightly different band positions for the same material [20].
- **Band overlap.** The Fe-O region is congested; apparent band positions shift with particle size, oxidation state and sampling geometry, which is why FTIR cannot settle the magnetite/maghemite question [10, 22].
- **UV-Vis.** As set out in Section 6, no band gap is reported, and dispersion optical data should not be treated as intrinsic electronic properties.
- **Particle statistics.**  $n = 84$  constrains the mean reasonably but not the distribution shape or its tails, and cannot distinguish lognormal from normal.
- **TGA.** Mass change only; conclusions depend on atmosphere, heating rate, sample mass and packing, all assumed here [26, 27].

## 11. Conclusion

This project demonstrates a complete, reproducible, five-technique characterization workflow for iron oxide nanoparticles, implemented in Python and applied to clearly-labelled synthetic datasets. The workflow indexed the diffraction pattern to cubic  $\text{Fe}_3\text{O}_4$  and refined a lattice parameter of  $8.3962 \pm 0.0073 \text{ \AA}$ ; separated size from strain broadening to give a crystallite size of  $11.9 \pm 0.6 \text{ nm}$ ; assigned the vibrational bands of a hydrated, uncoated spinel surface while flagging one band as an atmospheric artefact and declining to assign another; characterised a right-skewed particle population of 84 features; and resolved and quantified three thermal mass-loss events totalling 4.65 %.

Two outcomes matter more than the numbers. First, the XRD and SEM size metrics disagree by a factor of 1.60, and the analysis explains why that is the expected result of measuring different quantities with different weightings rather than a discrepancy to be explained away. Second, the Tauc analysis is carried out in full and then explicitly not reported, because four defensible analysis choices span 0.64–3.00 eV on the same data. Knowing which numbers not to report is as much a part of analytical competence as computing them.

The pipeline is validated against the parameters used to generate the data: all 55 automated checks pass, and the crystallite size, lattice parameter, particle-size distribution and thermal step amplitudes are all recovered within their stated tolerances. The datasets, analysis code, figures and this report are reproducible end to end from a single random seed.

## 12. References

*Every DOI below was resolved against the Crossref API and the returned metadata checked against the entry. Two references carry no DOI because none was issued. Full BibTeX records are in references/references.bib.*

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